

***Imagine deploying a container..
and Kubernetes automatically
blocks it because it violates
security policy***

What is Kubernetes Security?

- Protecting Kubernetes clusters, workloads, containers, and configurations from security risks
- Goal:
- Prevent insecure deployments
- Enforce security standards
- Reduce attack surface

Why Kubernetes Security is Critical?

- Kubernetes manages:
- Containers
- Networking
- Secrets
- Access control
- One weak configuration can impact the entire cluster

Common Kubernetes Security Risks

- Running containers as root
- Privileged containers
- Missing resource limits
- Open network communication
- Exposed secrets
- Insecure RBAC permissions

What is Policy-as-Code

- Security rules written as code
- Kubernetes automatically validates deployments against policies
- “Security enforcement becomes automated”

Why Policy-as-Code is Important?

- Manual reviews do not scale in large environments
- Policies provide:
 - Consistency
 - Automation
 - Governance
- Compliance enforcement

Real-World Example Policies

- Containers must not run as root
- Resource limits must exist
- Only approved images allowed
- Privileged mode must be blocked

Popular Kubernetes Policy Engines

Typical Kubernetes Security Workflow

Real-World Importance